SimpliCI

An AI-Based Intelligent Platform for Simplifying Complex Information

Project Proposal | CSE 4104 Software Development III | Week 02 Submission

Team Name	CSE4104-7D-T01	**Section**	7D
Team Leader	Utsho Kumar Dey (ID: 11230321321)	**Instructor**	Md. Riaz Mahmud, Assistant Professor
GitHub Repository	https://github.com/UTSHO2628/cse4104-7d-t01-simplici	**Department**	CSE, Northern University of Business and Technology, Khulna

Members: Utsho Kumar Dey (11230321321), Md. Adnun Ahemed Methun (11230321315), Madhurjoo Dam (11230321347), Mst Amena Akter Akhi (11230321322). Full details are in the Team Information Sheet (CSE4104-7D-T01_TeamInfo.pdf).

1. Project Title

SimpliCI: An AI-Based Intelligent Platform for Simplifying Complex Information

SimpliCI is a web platform that turns complex academic, medical and official documents into simple Bangla or English explanations at the reader's own level. The reader can then ask questions that are answered only from the document, and the system checks every answer against the source text before showing it.

2. Problem Statement

2.1 What problem currently exists

Students, patients and ordinary citizens regularly receive documents they cannot easily understand: research papers and textbooks, medical reports, government policies, circulars and administrative notices. These documents are long, full of technical terms and usually written in formal English. In Bangladesh the problem is larger because many readers are more comfortable in Bangla, and few tools explain complex text in simple Bangla at the reader's own level. As a result people skip important details, depend on other people to explain the document, or make decisions without fully understanding it.

2.2 Why the problem is important

- **Education:** students spend hours decoding papers, lecture material and notices instead of learning from them.
- **Health:** patients and families receive reports full of terms they cannot interpret. Understanding the report is the first step to asking the doctor the right questions.
- **Civic life:** policies and notices decide deadlines, eligibility and obligations. Misreading them has a real cost.
- **Trust in AI:** a general chatbot can give a fluent but wrong answer. For medical or official text this is dangerous, so simplification must be grounded in the document and checkable by the reader.

2.3 Why existing solutions are insufficient

Several good tools exist. The gap is not that they are bad, but that none of them is built around **domain-aware simplification, selectable reading level, checked evidence and quality monitoring in one workflow**.

Existing system	What it does	Limitation relevant to SimpliCI
ChatPDF	Chat with an uploaded PDF; multilingual; page-referenced answers	General purpose. No domain presets, no reading-level control, citations are not checked by the system, no admin or feedback loop. Reported to have no OCR for scanned PDFs.
Google NotebookLM	Source-grounded notes and Q&A with inline citations; supports Bengali output	General research workspace. Level and domain depend on the user's own prompts. No built-in structured outputs such as action items or alerts, and no admin quality monitoring.
Adobe Acrobat AI Assistant	Summaries and Q&A inside PDFs	Commercial, document-centric product. Not built around Bangla reading levels or domain presets.
Explainpaper	Explains highlighted parts of research papers	Academic papers only. No medical or policy support, no chat grounding checks.
General chatbots (ChatGPT, Gemini)	Summarise pasted text	User must write the prompt every time. No fixed structured output, no source verification, no document history, no admin monitoring.

Note: limitations are based on public feature descriptions checked in November 2026 and will be re-verified in Week 03 as stated in the Week 01 submission.

2.4 Why AI is relevant

A rule-based or keyword system cannot simplify arbitrary documents. AI is the core of this system, and the system cannot work without it:

- **Language understanding:** a large language model can read long, domain-specific text and rewrite it in simpler words while keeping the meaning.
- **Structured extraction:** one call returns summary, explanation, glossary, action items and alerts in a fixed JSON format that the backend validates and stores.
- **Personalisation:** one prompt pipeline adapts the same document to the chosen language, level and domain.
- **Grounded question answering:** the model answers using only the document text and returns the evidence it used, so users do not have to read every page.
- **Multimodal input:** a vision-capable model reads scanned pages and images when text extraction is not possible.

3. Objectives

- Build a web platform that simplifies PDF files, images and pasted text into Bangla or English at a selectable reading level.
- Provide domain-aware pipelines (Academic, Medical Report, Policy or Notice) using specialised prompt templates and structured JSON output.
- Provide document-grounded chat with page citations, and a clear "not found in the document" answer when the document does not contain the answer.
- Reduce made-up answers by **verifying** the evidence quoted by the AI against the real document text in the backend.
- Help users act on a document by extracting deadlines, amounts, obligations and warnings.
- Collect user feedback and give admins visibility into usage and AI quality so prompts can be improved with data.
- Deliver a complete, tested, deployed and documented system (frontend, backend, database, authentication, AI, deployment) within the semester.

4. Proposed Features

Features are planned in three layers so that a working core is always protected. Priority: **Must** = required for the MVP, **Should** = planned differentiator, **Could** = only if time permits.

4.1 Core platform (MVP)

ID	Feature	Description	Priority
F1	Authentication	Register and login with JWT and bcrypt password hashing; two roles: User and Admin.	Must
F2	Document input	Upload PDF or image, or paste text. File type and size limits enforced.	Must
F3	Simplification modes	Summary, simple explanation, key terms glossary, action items.	Must
F4	Personalization	Output language (Bangla or English) and reading level (Beginner or University).	Must
F5	Domain presets	Academic, Medical Report, Policy or Notice, each with its own prompt template and JSON schema.	Must
F6	Chat with document	Answers only from the uploaded document with page citation (paragraph for pasted text); says "not found" otherwise.	Must
F7	History and feedback	Dashboard of past documents and results; thumbs up or down on every AI output.	Must
F8	Admin dashboard	Users, usage statistics and feedback overview.	Must

4.2 Differentiator features (what makes SimpliCI stand out)

ID	Feature	Description	Priority
W1	Evidence verification	The AI must return a short quote as evidence for every chat answer. The backend checks that the quote really exists on the cited page. Answers that fail the check are flagged or rejected. Verified answers show a "Verified from page N" badge.	Should
W2	Split-view reader	Original document on one side, simplified output on the other. Clicking a point in the simplified text jumps to and highlights the source passage.	Should
W3	Smart Alerts panel	Automatically lists deadlines, amounts, obligations and warnings. For medical reports it also lists "questions to ask your doctor" (explanation only, never diagnosis).	Should
W4	Hover glossary	Key terms are underlined in the simplified text and show a simple definition on hover or tap.	Should
W5	Read aloud	Listen to the simplified text in Bangla or English (browser speech synthesis first, AI voice only if time permits).	Could
W6	Export and share	Download the simplified result as a PDF; create a read-only share link with an expiry time.	Should
W7	Privacy controls	Private file storage, delete-my-document and delete-my-data buttons, clear notice about sensitive documents.	Should
W8	AI quality analytics	Admin charts for feedback rate by domain, mode and language, failed-verification rate, response time and token use. Every output stores its prompt version and model name.	Should

4.3 Stretch goal (one only, decided at the Week 09 checkpoint)

- **S1 Multi-document library search:** ask one question across all of a user's documents using pgvector retrieval-augmented generation (RAG).
- **S2 Quiz and viva practice:** generate practice questions with answers from a document for students.

4.4 Responsible AI

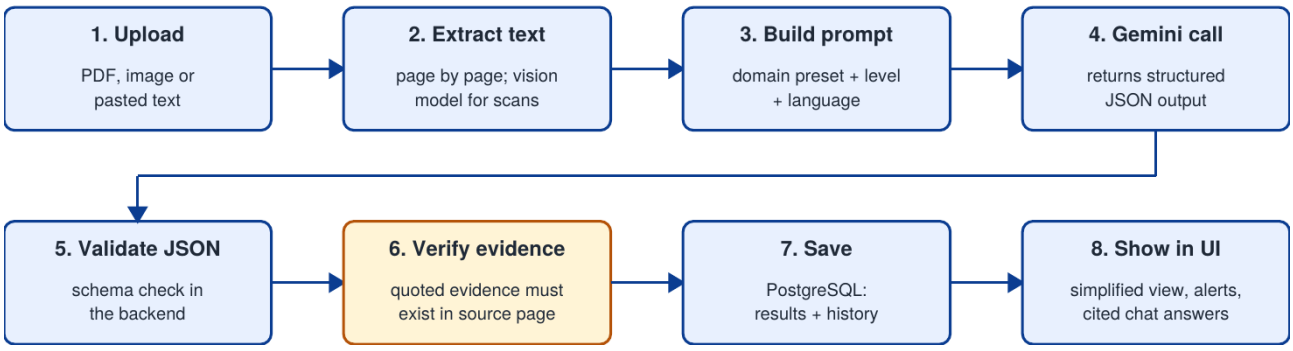
SimpliCI explains documents. It does not give medical, legal or financial advice, and every medical and policy result carries a visible reminder to confirm important decisions with a qualified person. Only dummy documents are used for development and testing, never real private or medical records.

5. Technology Stack

Layer	Technology	Purpose
Frontend	React (Vite), Tailwind CSS, react-pdf (pdf.js), Recharts	UI, split-view reader, dashboards and charts
Backend	Node.js + Express.js, Zod, Multer, Helmet, express-rate-limit	REST APIs, validation, file upload, security

Layer	Technology	Purpose
Database	PostgreSQL on Supabase (Postgres and file Storage only); pgvector if stretch S1 is chosen	Users, documents, results, chat, feedback
Authentication	Custom JWT with bcrypt password hashing, role-based access	Secure login, User and Admin roles
AI Integration	Google Gemini API (model name kept in GEMINI_MODEL environment variable); pdf-parse or pdfjs-dist for page-wise text; Gemini vision for scans	Simplification, structured JSON, grounded chat, OCR fallback
Deployment	Vercel (frontend), Render (backend), Supabase (database and storage)	Live public deployment on free tiers
Tools	GitHub, Postman, Figma, draw.io, dbdiagram.io, GitHub Projects	Version control, API testing, design, planning

AI workflow (the highlighted orange box is the step that separates SimpliCI from a plain chatbot wrapper):



Planned main tables: users, documents, document_pages, simplifications, chat_messages, feedback, share_links. Detailed design follows in Week 04.

Technology flexibility. As stated in Week 01, the stack may change when features change, for example a different Gemini model, OCR tool or hosting service. Changes are decided by the team leader with the team and reported in the next weekly submission.

Development approach. AI-assisted coding tools may be used to speed up development. All code is reviewed, tested and committed by the member who owns that module, and every member will be able to explain their module and the AI workflow in the viva.

6. Expected Outcome

6.1 What users will gain

- Students understand a paper, lecture note or notice in minutes and can ask the document questions instead of reading every page.
- Patients and families get a plain-language explanation of a report and a list of questions to ask the doctor.
- Citizens see the deadlines, amounts and obligations in a policy or circular without hunting through formal text.
- Every answer can be traced to a page, so users can trust it or check it.

6.2 How the system improves the existing workflow

Today	With SimpliCI
Read the whole long document in formal English	Get summary, explanation, glossary and action items in Bangla or English in one view
Ask someone else or look up terms one by one	Hover on key terms and ask questions about the document
A chatbot may invent an answer	Answers come only from the document, with a checked citation, or "not found"
Deadlines and amounts are easy to miss	Smart Alerts panel lists them
No way to know whether the AI is reliable	Feedback loop and admin analytics show quality by domain, language and prompt version

6.3 Expected benefits of AI integration

- One prompt pipeline serves three domains, two languages and two reading levels without building separate systems.
- Structured JSON output makes results consistent, storable and easy to display.
- Evidence verification and "not found" behaviour reduce hallucination, which is the main risk of AI document tools.

6.4 Measurable targets

These are targets to be measured in Week 10 testing, not results already achieved.

Target	How it will be measured
Complete flow works on the deployed system by Week 11: upload, simplify, chat, history, admin	Live demo and Postman test collection
Test set of 30 dummy documents: 10 Academic, 10 Medical, 10 Policy or Notice, including at least 6 Bangla and 6 scanned	Documented in the testing report
At least 90 percent of answerable test questions return a verified citation	Automated check of evidence against source pages
At least 90 percent of unanswerable test questions return "not found"	Manual review of the test question list
Typical 10-page text PDF simplified in about 30 seconds or less (depends on free-tier limits)	Response time logged in the database
All items of the course Minimum Project Requirements checklist completed	Checklist in the Project Plan document

6.5 Feasibility and scope control

The core platform (F1 to F8) is built first and must be working by Week 09. Differentiators are added one by one after that. If the team is behind schedule at the Week 09 checkpoint, W2 (highlighting), W5 and the stretch goal are dropped first, so the project always ends complete, stable, tested and documented. The week-by-week plan, roles and risks are in CSE4104-7D-T01_ProjectPlan.pdf.